

Sequences and Series

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9.1 Sequences

Definition 9.1. A *sequence* in a set X is a function $a : \mathbb{N} \rightarrow X$ (or $\mathbb{Z}^+ \rightarrow X$); we write a_n for $a(n)$ and $(a_n)_{n \geq 0}$ for the sequence. A *subsequence* $(a_{n_k})_{k \geq 0}$ is obtained by selecting indices $n_0 < n_1 < n_2 < \dots$.

Definition 9.2. A real sequence (a_n) is *bounded* if $\exists M, |a_n| \leq M$ for all n ; *increasing* if $a_{n+1} \geq a_n$ for all n (strictly, with $>$); *decreasing* dually; *monotone* if either.

9.2 Limits

Definition 9.3 (Limit). (a_n) *converges* to $L \in \mathbb{R}$, written $\lim_{n \rightarrow \infty} a_n = L$, when

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N, |a_n - L| < \varepsilon.$$

A sequence with no limit *diverges*.

Example 9.4. $a_n = \frac{1}{n} \rightarrow 0$: given $\varepsilon > 0$ take $N > 1/\varepsilon$; then $n \geq N \Rightarrow |1/n| \leq 1/N < \varepsilon$.

Theorem 9.5 (Uniqueness). A sequence has at most one limit.

Proof. If $a_n \rightarrow L$ and $a_n \rightarrow L'$ with $L \neq L'$, take $\varepsilon = |L - L'|/2$. Beyond the larger of the two thresholds, $|L - L'| \leq |L - a_n| + |a_n - L'| < 2\varepsilon = |L - L'|$ — contradiction. $\square$

Theorem 9.6 (Algebra of limits). If $a_n \rightarrow A$ and $b_n \rightarrow B$ then $a_n + b_n \rightarrow A + B$, $a_n b_n \rightarrow AB$, and $a_n/b_n \rightarrow A/B$ provided $B \neq 0$.

Theorem 9.7 (Monotone convergence). Every increasing sequence bounded above converges (to its supremum); every decreasing sequence bounded below converges.

Theorem 9.8 (Squeeze). If $a_n \leq b_n \leq c_n$ eventually and $a_n \rightarrow L$, $c_n \rightarrow L$, then $b_n \rightarrow L$.

Definition 9.9. (a_n) is *Cauchy* when $\forall \varepsilon > 0 \exists N \forall m, n \geq N, |a_m - a_n| < \varepsilon$.

In $\mathbb{R}$, a sequence converges iff it is Cauchy (completeness), and every bounded sequence has a convergent subsequence (Bolzano–Weierstrass).

9.2.1 Standard limits

$$\frac{1}{n^p} \rightarrow 0 \ (p > 0); \quad r^n \rightarrow 0 \ (|r| < 1); \quad n^{1/n} \rightarrow 1; \quad \left(1 + \frac{x}{n}\right)^n \rightarrow e^x; \quad \frac{n!}{n^n} \rightarrow 0.$$

9.3 Progressions and finite sums

Definition 9.10 (AP). An *arithmetic progression* has constant difference: $a_n = a_0 + nd$. The sum of the first n terms is

$$S_n = \sum_{k=0}^{n-1} (a_0 + kd) = na_0 + d \frac{n(n-1)}{2} = \frac{n}{2} (\text{first} + \text{last}).$$

Definition 9.11 (GP). A *geometric progression* has constant ratio: $a_n = a_0 r^n$. For $r \neq 1$,

$$\sum_{k=0}^{n-1} a_0 r^k = a_0 \frac{1 - r^n}{1 - r}, \quad \text{and for } |r| < 1, \quad \sum_{k=0}^{\infty} a_0 r^k = \frac{a_0}{1 - r}.$$

Theorem 9.12 (Standard closed forms).

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Proof of the second formula, by induction. Base $n = 1$: $1 = \frac{1 \cdot 2 \cdot 3}{6}$. Step: assuming the formula for n ,

$$\sum_{k=1}^{n+1} k^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)(2n^2 + n + 6n + 6)}{6} = \frac{(n+1)(n+2)(2n+3)}{6},$$

which is the formula at $n+1$. (Induction is treated fully in Chapter 13.) $\square$

9.4 Series

Definition 9.13. Given $(a_n)_{n \geq 1}$, the *series* $\sum_{n \geq 1} a_n$ is the sequence of *partial sums* $S_N = \sum_{n=1}^N a_n$; the series *converges* to S when $S_N \rightarrow S$.

Theorem 9.14 (Necessary condition). If $\sum a_n$ converges then $a_n \rightarrow 0$. The converse fails: the *harmonic series* $\sum 1/n$ diverges although $1/n \rightarrow 0$.

Divergence of the harmonic series. Group terms: $\frac{1}{2} \geq \frac{1}{2}$; $\frac{1}{3} + \frac{1}{4} > \frac{2}{4} = \frac{1}{2}$; $\frac{1}{5} + \dots + \frac{1}{8} > \frac{4}{8} = \frac{1}{2}$; each block of 2^k terms exceeds $\frac{1}{2}$, so $S_{2^m} > 1 + \frac{m}{2} \rightarrow \infty$. $\square$

Example 9.15 (Telescoping). $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{N+1} \right) = 1$.

Theorem 9.16 (p -series). $\sum_{n \geq 1} n^{-p}$ converges iff $p > 1$.

9.4.1 Convergence tests

Theorem 9.17 (Tests for positive series). Let $a_n, b_n \geq 0$.

- **Comparison:** if $a_n \leq b_n$ eventually and $\sum b_n$ converges, then $\sum a_n$ converges; contrapositively divergence passes upward.
- **Ratio (d'Alembert):** if $a_{n+1}/a_n \rightarrow \ell$, the series converges for $\ell < 1$ and diverges for $\ell > 1$ ($\ell = 1$: inconclusive).
- **Root (Cauchy):** same conclusion with $\sqrt[n]{a_n} \rightarrow \ell$.
- **Integral:** for f positive decreasing, $\sum f(n)$ and $\int_1^{\infty} f(x) dx$ converge or diverge together.

Theorem 9.18 (Leibniz alternating test). If (b_n) decreases to 0, then $\sum (-1)^{n+1} b_n$ converges, and the error after N terms is at most b_{N+1} .

Definition 9.19. $\sum a_n$ is *absolutely convergent* when $\sum |a_n|$ converges; absolute convergence implies convergence. A convergent but not absolutely convergent series (e.g. $\sum (-1)^{n+1}/n$) is *conditionally convergent* — and by Riemann's theorem its terms can be rearranged to reach any prescribed sum.

9.5 Recurrence relations

Definition 9.20. A sequence is defined by a *recurrence* when each term is given as a function of earlier terms, together with initial conditions — e.g. the Fibonacci sequence $F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2}$.

Theorem 9.21 (Linear homogeneous recurrences, order 2). For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, form the characteristic equation $x^2 = c_1 x + c_2$. If it has distinct roots $r_1 \neq r_2$, the general solution is $a_n = \alpha r_1^n + \beta r_2^n$; if a double root r , then $a_n = (\alpha + \beta n) r^n$. The constants are fixed by the initial conditions.

Example 9.22 (Binet's formula). For Fibonacci, $x^2 = x + 1$ gives roots $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$. Solving $\alpha + \beta = 0$, $\alpha\varphi + \beta\psi = 1$ yields

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

Since $|\psi| < 1$, F_n is the nearest integer to $\varphi^n/\sqrt{5}$, and $F_{n+1}/F_n \rightarrow \varphi$, the golden ratio.

9.6 Products and famous sequences

The product notation mirrors the sum: $\prod_{k=1}^n a_k = a_1 a_2 \cdots a_n$; e.g. $n! = \prod_{k=1}^n k$, and Wallis' product $\prod_{k \geq 1} \frac{4k^2}{4k^2 - 1} = \frac{\pi}{2}$. Beyond Fibonacci, classical sequences include the Catalan numbers $C_n = \frac{1}{n+1} \binom{2n}{n}$ (Chapter 11), the harmonic numbers $H_n = \sum_{k \leq n} 1/k \approx \ln n + \gamma$, the primes, and the triangular numbers $n(n+1)/2$.

9.7 Summary

Sequences are functions on $\mathbb{N}$; convergence is the ε - N game, tamed by uniqueness, limit algebra, monotone convergence, squeeze and Cauchy criteria. AP/GP sums and the power sums of Theorem 9.12 are the basic closed forms. Series converge through comparison, ratio, root, integral and Leibniz tests; absolute convergence is the robust notion. Linear recurrences are solved by characteristic roots — Binet's formula being the showcase.

Exercises

Exercise 9.1 EASY

Write the first five terms of: (a) $a_n = \frac{(-1)^n}{n+1}$ ($n \geq 0$); (b) the sequence defined by $b_0 = 2$, $b_n = 3b_{n-1} - 1$.

Exercise 9.2 EASY

An amphitheatre has 20 rows; the first has 18 seats and each row has 3 more than the previous. How many seats in total?

Exercise 9.3 EASY

Compute $\sum_{k=0}^9 5 \cdot 2^k$ and $\sum_{k=0}^{\infty} 5 \cdot (1/3)^k$.

Exercise 9.4 MEDIUM

Prove from Definition 9.3 that $\lim_{n \rightarrow \infty} \frac{2n+1}{n+3} = 2$.

Exercise 9.5 MEDIUM

Let $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$. Show (a_n) is increasing and bounded above by 2, and find its limit.

Exercise 9.6 MEDIUM

Using Theorem 9.12, find a closed form for $\sum_{k=1}^n (2k-1)$ (the sum of the first n odd numbers) and for $\sum_{k=1}^n k(k+1)$.

Exercise 9.7 MEDIUM

Determine the convergence of: (a) $\sum \frac{n}{2^n}$; (b) $\sum \frac{1}{\sqrt{n}}$; (c) $\sum \frac{(-1)^n}{\ln(n+1)}$.

Exercise 9.8 MEDIUM

Evaluate the telescoping series $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}$.

Exercise 9.9 MEDIUM

Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 1$, $a_1 = 0$.

Exercise 9.10 HARD

Solve $a_n = 4a_{n-1} - 4a_{n-2}$, $a_0 = 1$, $a_1 = 4$ (double root case).

Exercise 9.11 HARD

Prove by induction the bound $F_n \leq \varphi^{n-1}$ for $n \geq 1$, where $\varphi = \frac{1+\sqrt{5}}{2}$ (note $\varphi^2 = \varphi + 1$).

Exercise 9.12 HARD

Show that $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$ diverges although $\frac{1}{n \ln n} \rightarrow 0$, using the integral test.

Solutions to the Exercises

Solution 9.1. (a) $1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5}$. (b) 2, 5, 14, 41, 122. ■

Solution 9.2. AP with $a_1 = 18$, $d = 3$, $n = 20$: last row has $18 + 19 \cdot 3 = 75$ seats; total $\frac{20}{2}(18 + 75) = 10 \cdot 93 = 930$. ■

Solution 9.3. $5 \frac{2^{10}-1}{2-1} = 5 \cdot 1023 = 5115$; $\frac{5}{1-1/3} = \frac{15}{2}$. ■

Solution 9.4. $\left| \frac{2n+1}{n+3} - 2 \right| = \frac{5}{n+3}$. Given $\varepsilon > 0$, choose $N > \frac{5}{\varepsilon} - 3$; then $n \geq N \Rightarrow \frac{5}{n+3} \leq \frac{5}{N+3} < \varepsilon$. ■

Solution 9.5. Bounded: by induction, $a_n < 2 \Rightarrow a_{n+1} = \sqrt{2+a_n} < \sqrt{4} = 2$, and $a_1 = \sqrt{2} < 2$. Increasing: $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(1 + a_n) > 0$. By monotone convergence the limit L exists and satisfies $L = \sqrt{2+L}$, i.e. $L^2 - L - 2 = 0$, so $L = 2$ (the root -1 being impossible). ■

Solution 9.6. $\sum_{k=1}^n (2k-1) = 2 \frac{n(n+1)}{2} - n = n^2$. $\sum k(k+1) = \sum k^2 + \sum k = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)(n+2)}{3}$. ■

Solution 9.7. (a) Ratio test: $\frac{(n+1)/2^{n+1}}{n/2^n} = \frac{n+1}{2n} \rightarrow \frac{1}{2} < 1$: converges. (b) p -series with $p = \frac{1}{2} \leq 1$: diverges. (c) $\frac{1}{\ln(n+1)}$ decreases to 0: converges by Leibniz (only conditionally — compare with the divergent $\sum 1/(n \ln n)$ -type behaviour of the absolute series, which dominates $1/n$ eventually... in fact $\frac{1}{\ln(n+1)} > \frac{1}{n}$ for $n \geq 1$, so absolute convergence fails by comparison with the harmonic series). ■

Solution 9.8. $\frac{1}{(2n-1)(2n+1)} = \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$, so the partial sum is $\frac{1}{2} \left(1 - \frac{1}{2N+1} \right) \rightarrow \frac{1}{2}$. ■

Solution 9.9. Characteristic equation $x^2 = 5x - 6$: roots 2, 3. General solution $a_n = \alpha 2^n + \beta 3^n$; conditions give $\alpha + \beta = 1$, $2\alpha + 3\beta = 0$, so $\beta = -2$, $\alpha = 3$: $a_n = 3 \cdot 2^n - 2 \cdot 3^n$. ■

Solution 9.10. $x^2 = 4x - 4$ has the double root $r = 2$, so $a_n = (\alpha + \beta n)2^n$. From $a_0 = 1$: $\alpha = 1$; from $a_1 = 4$: $(1 + \beta) \cdot 2 = 4$, $\beta = 1$. Hence $a_n = (n + 1) 2^n$. ■

Solution 9.11. Strong induction. Base: $F_1 = 1 = \varphi^0$ and $F_2 = 1 \leq \varphi^1$. Step ($n \geq 3$): assuming the bound below n ,

$$F_n = F_{n-1} + F_{n-2} \leq \varphi^{n-2} + \varphi^{n-3} = \varphi^{n-3}(\varphi + 1) = \varphi^{n-3}\varphi^2 = \varphi^{n-1}.$$

■

Solution 9.12. $f(x) = \frac{1}{x \ln x}$ is positive and decreasing on $[2, \infty)$, and $\int_2^T \frac{dx}{x \ln x} = \ln \ln T - \ln \ln 2 \rightarrow \infty$. By the integral test the series diverges — terms tending to 0 is necessary but never sufficient (Theorem 9.14). ■